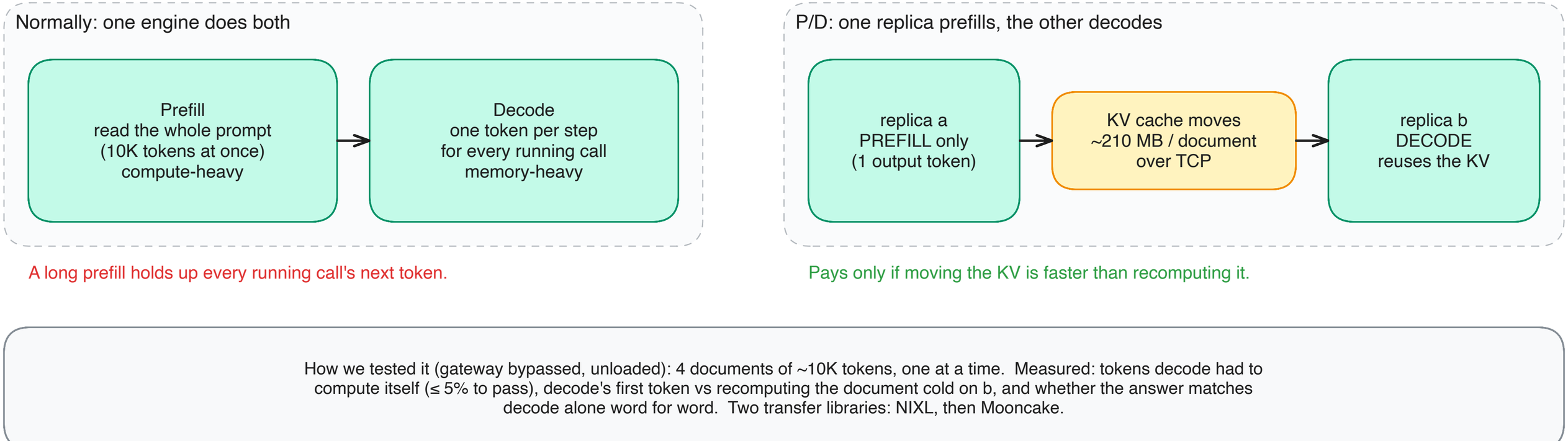

Normally: one engine does both
Prefill
read the whole prompt
(10K tokens at once)
compute-heavy
Decode
one token per step
for every running call
memory-heavy
A long prefill holds up every running call's next token.
P/D: one replica prefills, the other decodes
replica a
PREFILL only
(1 output token)
KV cache moves
~210 MB / document
over TCP
replica b
DECODE
reuses the KV
Pays only if moving the KV is faster than recomputing it.
How we tested it (gateway bypassed, unloaded): 4 documents of ~10K tokens, one at a time. Measured: tokens decode had to compute itself (≤ 5% to pass), decode's first token vs recomputing the document cold on b, and whether the answer matches decode alone word for word. Two transfer libraries: NIXL, then Mooncake.